TryHackMe_Linux_Fundamentals_Pt1.md

TryHackMe: Linux Fundamentals — Technical Write-Up

Module Overview

This room introduces essential Linux operating system concepts, including system navigation via the command-line interface, file searching utilities, and basic terminal management.

Task 1: Introduction to Linux

Concept: Linux powers a vast array of global infrastructure, ranging from enterprise web servers and car entertainment systems to point-of-sale (PoS) terminals and critical industrial infrastructure.

Practical Workflow:

Deployed the interactive browser-based Ubuntu Linux machine by clicking the Start Machine button.
Verified that the target IP address and active terminal environment were properly initialized.

Task 3: Finding Your Way Around (Navigation)

Concept: Learning how to navigate file systems using core terminal commands without relying on a graphical user interface (GUI) mouse.

ls: Lists the contents of the current working directory.
cd: Changes the directory path.
cat: Outputs and displays the contents of a text file.
pwd: Prints the absolute path of the current working directory.

Practical Workflow:

Ran the ls command in the home folder to count the available directories.
Explored subdirectories using cd and utilized ls to locate the folder containing the target text file.
Read the file contents using the cat command.

Questions & Answers:

Question 1: Run ls in the current folder. How many folders are there? \rightarrow 4
Question 2: One of those folders contains a file. Which folder is it? \rightarrow (Specific folder name discovered during the lab session)

Question 3: Use cat to read the file within this folder. What does it say? \rightarrow (Specific file content string/flag discovered during the lab session)

Task 4: Let the Machine Do the Searching

Concept: Using advanced command-line tools to search efficiently through file structures and text logs instead of manually reading massive text blocks.

find: Locates files by name (e.g., find -name passwords.txt).
grep: Searches inside files for specific text strings or patterns (e.g., grep "password123" passwords.txt).

Practical Workflow:

Navigated back to the home directory (cd /home/tryhackme).
Executed the grep utility to filter and extract a specific flag string from a large web server log file (access.log).